NETWORKING

UDP Visual Guide

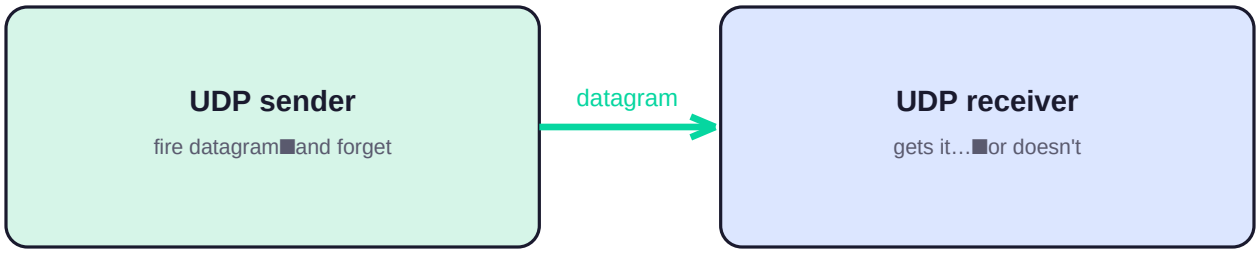
Connectionless datagrams —
fast, simple, and honest about it.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Connectionless by Design

No handshake, no session, no promises.

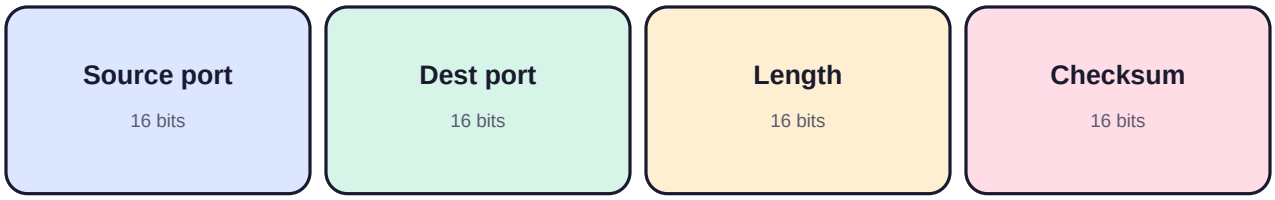


- No connection setup: the first packet IS the data — zero round-trip overhead.
- No ordering, no retransmission, no flow control: packets may arrive late, duplicated, or never.
- The app takes responsibility for anything it actually needs (retries, sequencing) — or doesn't need it at all.

STEP 2/5

The 8-Byte Header

The smallest transport header in common use — TCP's is 20+ bytes.



- That's the ENTIRE header: two ports, total length, checksum. Nothing else.
- No sequence numbers, no ACKs, no window — the fields TCP needs simply don't exist.
- Checksum is optional in IPv4 (often zeroed) but mandatory in IPv6.

STEP 3/5

Where UDP Wins

Real protocols that chose speed and simplicity.

DNS

One question, one answer — a handshake would double the latency.

Gaming & VoIP

A late packet is useless; drop it and move on. Jitter buffers smooth arrival.

Live video (RTP)

Losing a frame beats freezing the stream waiting for a retransmit.

QUIC / HTTP/3

Reliability rebuilt in user space on top of UDP — the best of both.

STEP 4/5

UDP vs TCP: The Decision

Pick TCP when

correctness first:■files, web APIs,■email, DB

Pick UDP when

timeliness first:■DNS, voice, games,■live media

- Can you tolerate loss? Yes → UDP is on the table. No → TCP.
- Do stale packets hurt more than missing ones? (voice, games) → UDP.
- Want TCP reliability without TCP's head-of-line blocking? → QUIC over UDP.

STEP 5/5

UDP's Sharp Edges

Know the trade-offs before you commit.

- No congestion control: a UDP flood doesn't back off — it can starve TCP flows on the same link.
- Middlebox trouble: some firewalls and NATs treat UDP harshly; idle mappings expire fast (keep-alives needed).
- Amplification abuse: spoofed-source UDP services (DNS, NTP) power DDoS reflection — run with care, rate-limit responses.
- MTU matters: oversized datagrams fragment, and one lost fragment kills the whole datagram — stay under ~1400 bytes.

Rule of thumb: UDP is a toolkit, not a transport with opinions. The reliability you skip must be consciously skipped — or rebuilt above it.

STEP ✓/✓

Key Takeaways

UDP — Remember

- Connectionless: no handshake, no ordering, no retransmission — first packet is data.
- 8-byte header: source/dest ports, length, checksum. That's everything.
- Wins where timeliness beats completeness: DNS, gaming, VoIP, live video, QUIC.
- Choose UDP when stale data is worse than missing data.
- Sharp edges: no congestion control, NAT timeouts, amplification abuse, fragmentation.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev